Part III — Grading Rubric

Problem Solving (60 pts) + Bonus (3 pts). g = 10 m/s². Right / up / east positive. Scoring = method + answer split.

■ PARTIAL CREDIT: award method marks even if the final number is wrong. Green badges show where each point goes.

Q1 · Forces on a block (6 pts)

Given: m = 4 kg, F■ = +18 N (right), F■ = –6 N (left), frictionless.

a) Net force (2 pts)

Fnet = F■ + F■ = (+18) + (–6) = +12 N (to the right)

Setup Fnet = F■ + F■ = 1 pt	**+12 N with direction = 1 pt**

b) Acceleration (2 pts)

a = Fnet / m = 12 / 4 = +3 m/s² (to the right)

Setup a = Fnet/m = 1 pt	**+3 m/s² with direction = 1 pt**

c) Balanced or unbalanced? (2 pts)

Unbalanced — Fnet ≠ 0 (Fnet = +12 N), so the block accelerates right.

States 'unbalanced' = 1 pt	**Justifies: Fnet ≠ 0 = 1 pt**

Q2 · Suitcase in an elevator (6 pts)

Given: m = 25 kg, g = 10 m/s², elevator at rest.

a) Weight Fg (3 pts)

Fg = m·g = 25 × 10 = 250 N (downward)

Formula Fg = m·g = 1 pt	**Substitute 25 × 10 = 1 pt**	**250 N downward = 1 pt**

b) Normal force Fn (3 pts)

At rest a = 0 → Fnet = 0 → Fn – Fg = 0 → Fn = Fg

Fn = 250 N (upward)

Reasoning a = 0 → Fn = Fg = 1 pt	**Setup Fn – Fg = 0 = 1 pt**	**250 N upward = 1 pt**

Q3 · Worker pulls a toolbox (5 pts)

Given: F_pull = 50 N (horizontal), f = 15 N (opposes), d = 20 m.

a) Work by the pull (2 pts)

Pull ■ displacement → $\theta = 0°$, $\cos 0 = 1$

$W\_pull = F \cdot d \cdot \cos 0 = 50 \times 20 \times 1 = +1000$ J

Uses cos0 = 1 (W = F·d·cosθ) = 1 pt	**+1000 J = 1 pt**

b) Work by friction (2 pts)

Friction opposes → $\theta = 180°$, $\cos 180 = -1$

$W\_f = f \cdot d \cdot \cos 180 = -(15)(20) = -300$ J

Uses cos180 = –1 (negative work) = 1 pt	**–300 J = 1 pt**

c) Net work (1 pt)

$W\_net = W\_pull + W\_f = (+1000) + (-300) = +700$ J

+700 J = 1 pt

Q4 · Work–energy theorem (6 pts)

Given: m = 3 kg, v■ = 4 m/s, W_net = +30 J.

a) Initial KE■ (3 pts)

$KE\blacksquare = \tfrac{1}{2} \cdot m \cdot v\blacksquare^2 = \tfrac{1}{2} \times 3 \times 4^2 = \tfrac{1}{2} \times 3 \times 16 = 24$ J

Formula KE = ½mv² = 1 pt	**Substitute ½×3×16 = 1 pt**	**24 J = 1 pt**

b) Final KE_f via W_net = ΔKE (3 pts)

$W\_net = KE\_f - KE\blacksquare \rightarrow KE\_f = W\_net + KE\blacksquare = 30 + 24 = 54$ J

Uses W_net = ΔKE = 1 pt	**Substitute 30 + 24 = 1 pt**	**54 J = 1 pt**

Q5 · Energy of a falling ball (6 pts)

Given: m = 0.5 kg, h = 5 m, g = 10, released from rest (v■ = 0), ground h = 0.

a) PEg at release + state KE (3 pts)

$PEg = m \cdot g \cdot h = 0.5 \times 10 \times 5 = 25$ J

At release $v\blacksquare = 0 \rightarrow KE = 0$ J

Formula PEg = mgh = 1 pt	**25 J = 1 pt**	**KE = 0 J at release = 1 pt**

b) Total mechanical energy ME (3 pts)

$ME = KE + PEg = 0 + 25 = 25$ J

Uses ME = KE + PEg = 1 pt	**Substitute 0 + 25 = 1 pt**	**25 J = 1 pt**

Q6 · Momentum of a ball (5 pts)

Given: m = 2 kg, v■ = +8 m/s (right), v■ = –8 m/s (left); right positive.

a) Momentum p■ (2 pts)

p■ = m·v■ = 2 × (+8) = +16 kg·m/s (to the right)

Setup p = m·v = 1 pt	**+16 kg·m/s right = 1 pt**

b) New momentum p■ (2 pts)

p■ = m·v■ = 2 × (–8) = –16 kg·m/s (to the left)

Setup p = m·v = 1 pt	**–16 kg·m/s left = 1 pt**

c) Change in momentum Δp (1 pt)

Δp = p■ – p■ = (–16) – (+16) = –32 kg·m/s (to the left)

–32 kg·m/s = 1 pt

Q7 · Impulse on a cart (6 pts)

Given: F = +40 N, m = 5 kg, v■ = 0 (at rest), Δt = 0.5 s, frictionless.

a) Impulse J (2 pts)

J = F·Δt = 40 × 0.5 = +20 N·s

Setup J = F·Δt = 1 pt	**+20 N·s = 1 pt**

b) Initial momentum p■ (2 pts)

p■ = m·v■ = 5 × 0 = 0 kg·m/s (starts at rest)

Setup p■ = m·v■ = 1 pt	**0 kg·m/s = 1 pt**

c) Final momentum p_f via J = Δp (2 pts)

p_f = p■ + J = 0 + 20 = +20 kg·m/s (direction of force)

Uses J = Δp = p_f – p■ = 1 pt	**+20 kg·m/s = 1 pt**

Q8 · Baseball struck by a bat (6 pts)

Given: m = 0.15 kg, v■ = +30 m/s, v_f = –20 m/s, Δt = 0.02 s.

a) Initial & final momenta (2 pts)

p■ = m·v■ = 0.15 × (+30) = +4.5 kg·m/s

p_f = m·v_f = 0.15 × (–20) = –3.0 kg·m/s

Setup p = m·v = 1 pt	**+4.5 and –3.0 kg·m/s = 1 pt**

b) Change in momentum Δp (2 pts)

Δp = p_f – p■ = (–3.0) – (+4.5) = –7.5 kg·m/s

Setup Δp = p_f – p■ = 1 pt	**–7.5 kg·m/s = 1 pt**

c) Force from the bat (2 pts)

F = Δp / Δt = (–7.5) / (0.02) = –375 N (opposite to initial motion)

Uses F = Δp/Δt = 1 pt	**–375 N = 1 pt**

Q9 · Two skaters push off (7 pts)

Given: m■ = 60 kg (rest), m■ = 40 kg (rest); after push v■f = –2 m/s. Left –, right +.

a) Total momentum before push (2 pts)

Both at rest → p_total,i = 0 kg·m/s (system initially at rest)

p_total,i = 0 = 1 pt	**Justifies: both at rest = 1 pt**

b) Velocity of 40 kg skater after push (5 pts)

0 = m■·v■f + m■·v■f

0 = (60)(–2) + (40)·v■f = –120 + 40·v■f

v■f = +120 / 40 = +3 m/s (to the right — opposite the 60 kg skater)

Conservation: 0 = m■v■f + m■v■f = 2 pt	**Substitute values = 1 pt**	**Solve for v■f = 1 pt**	**+3 m/s with direction = 1 pt**

Q10 · Car collision — perfectly inelastic (7 pts)

Given: m_A = 1200 kg, v_Ai = +25 m/s (east); m_B = 1800 kg, v_Bi = 0; cars lock.

a) Total momentum before (2 pts)

p_total,i = m_A·v_Ai + m_B·v_Bi = (1200)(+25) + (1800)(0)

p_total,i = +30,000 kg·m/s (east)

Setup p = m_Av_A + m_Bv_B = 1 pt	**+30,000 kg·m/s east = 1 pt**

b) Common velocity after (5 pts)

p_before = (m_A + m_B)·v_f → +30,000 = 3000·v_f

v_f = 30,000 / 3000 = +10 m/s (east)

Inelastic: p_i = (m_A+m_B)v_f = 2 pt	**Substitute 30,000 = 3000·v_f = 1 pt**	**Solve = 1 pt**	**+10 m/s east = 1 pt**

BONUS · Braking with impulse (3 pts — Extra Credit)

Given: m = 1500 kg, v■ = +20 m/s, v_f = 0, F = –300 N.

Time to stop (3 pts)

J = Δp = m·v_f – m·v■ = (1500)(0) – (1500)(+20) = –30,000 kg·m/s

J = F·Δt → Δt = J / F = (–30,000) / (–300) = 100 s

J = Δp = mv_f – mv■ = 1 pt	**J = –30,000 kg·m/s = 1 pt**	**Δt = 100 s = 1 pt**